Practice Midterm Questions

CS/STAT 184(0): Introduction to Reinforcement Learning

Here are some midterm practice questions. The first three questions comprised the midterm two years ago, and the next three last year. Note that these were challenging exams, and effort is being made to reduce the length this year. We recommend trying the questions on your own before looking at the solutions.

1 Regret of Bandit Algorithms [18 points]

In this section, we are going to study the regret of some bandit algorithms that are distinct from, but have a similar flavor to, the UCB algorithm. But this problem is not about formal rigorous proofs, so we are just asking you to determine whether each algorithm's regret is linear or sub-linear, and just convincing explanations in words will suffice to get full credit.

Recall the uniform Hoeffding bound we derived in lecture 3:

$$\mathbb{P}\left(\forall k \leq K, t < T,\ |\hat{\mu}_t^{(k)} - \mu^{(k)}| \leq \sqrt{\ln(2TK/\delta)/2N_t^{(k)}}\right) \geq 1 - \delta. \tag{1}$$

and that this bound can be used to define uniform confidence intervals on $\mu^{(k)}$.

Recall that the UCB algorithm chooses the arm at time t that maximizes the upper endpoint of the uniform confidence interval implied by equation (1):

$$a_t^{(\text{UCB})} := \arg\max_{k \in K} \left\{ \hat{\mu}_t^{(k)} + \sqrt{\ln(2TK/\delta)/2N_t^{(k)}} \right\}.$$

Define $\hat{S}_t^*$ as the set of arms at time t whose confidence intervals overlap the confidence interval of the arm with the highest upper endpoint (which is $a_t^{(\text{UCB})}$), i.e.,

$$\hat{S}_t^* = \left\{ k : \hat{\mu}_t^{(k)} + \sqrt{\ln(2TK/\delta)/2N_t^{(k)}} \geq \hat{\mu}_t^{(a_t^{(\text{UCB})})} - \sqrt{\ln(2TK/\delta)/2N_t^{(a_t^{(\text{UCB})})}} \right\}.$$

1.1 A different UCB algorithm [6 points]

Note that UCB can equivalently be defined as choosing the arm with the highest UCB among $\hat{S}_t^*$. Consider the algorithm that instead chooses the arm with the lowest UCB among $\hat{S}_t^*$:

$$a_t^{(\mathrm{UCB}^*)} := \arg\min_{k \in \hat{S}_t^*} \left\{ \hat{\mu}_t^{(k)} + \sqrt{\ln(2TK/\delta)/2N_t^{(k)}} \right\}.$$

Does this algorithm have sublinear regret?

1.2 A lower confidence bound (LCB) algorithm [6 points]

Consider the algorithm that chooses the arm with the lowest LCB:

$$a_t^{(\mathrm{LCB})} := \arg\min_{k \in K} \left\{ \hat{\mu}_t^{(k)} - \sqrt{\ln(2TK/\delta)/2N_t^{(k)}} \right\}.$$

Does this algorithm have sublinear regret?

1.3 A different LCB algorithm [6 points]

Consider the algorithm that chooses the arm with the lowest LCB within $\hat{S}_t^*$:

$$a_t^{(\mathrm{LCB}*)} := \arg\min_{k \in \hat{S}_t^*} \left\{ \hat{\mu}_t^{(k)} - \sqrt{\ln(2TK/\delta)/2N_t^{(k)}} \right\}.$$

Does this algorithm have sublinear regret?

2 Tabular MDP [20 points]

Consider a simple MDP with two states $\mathcal{S} = \{s_0, s_1\}$, two actions $\mathcal{A} = \{a_0, a_1\}$, and discount factor $\gamma \in (0, 1)$. States and actions contribute to the overall reward independently: Being in s_0 contributes no value while being in s_1 corresponds to a value of 1 reward unit. Taking action a_0 costs nothing while a_1 costs $\frac{1}{2}$ reward units. Once in s_1 the system always stays in s_1, regardless of the action taken. From state s_0, taking action a_1 causes a deterministic transition to s_1, while under action a_0 the system will likely stay in s_0, but might transition to s_1 with probability ϵ.

2.1 3 points

Write the transition probabilities $P(s'|s, a)$ and the rewards $r(s, a)$ for all s', a, s.

2.2 3 points

What is the optimal action in s_1, i.e., what is $\pi^*(s_1)$? Justify your answer.

2.3 7 points

Consider two policies π_0 and π_1 defined as $\pi_0(s_1) = \pi_1(s_1) = \pi^*(s_1)$ and

$$\pi_0(s_0) = a_0, \quad \pi_1(s_0) = a_1 \ .$$

Write their value functions.

2.4 7 points

Find a threshold ϵ_0 such that when $\epsilon \leq \epsilon_0$, the policy π_1 is optimal. For which values of γ is this a valid probability threshold (i.e. $\epsilon_0 > 0$)?

3 Infinite horizon linear quadratic regulator (LQR) [10 points]

In this section, we are going to study the infinite horizon discounted LQR problem, but to keep things a bit simpler, we will assume the state and action dimensions are 1-dimensional. That is, the dynamics are given by

$$s_{t+1} = As_t + Ba_t + w_t, \qquad s_0 \sim \mu_0, \qquad w_t \sim \mathcal{N}(0, \sigma^2),$$

the cost function is given by

$$c(s, a) = Qs^2 + Ra^2,$$

and the objective is to minimize the infinite horizon discounted reward with discount factor $\gamma \in (0, 1)$:

$$\mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t)\right],$$

where all variables appearing the above equations in this section are scalars.

3.1 10 points

Show that for a linear policy, i.e., a policy of the form

$$a_t = ms_t$$

for some scalar m, the value function is quadratic, i.e., given by $cs^2 + d$, for some scalars c and d, and find the values of c and d in terms of the policy and problem parameters. Hint: use Bellman consistency equations ($V^\pi(s_t) = c(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim f(A,B,w_{t+1})}[V^\pi(s_{t+1})]$). You may use without proof that they have a unique solution.

4 Advantages [35 points]

Consider a discounted MDP, $\mathcal{M}_\gamma = \{\mu, S, A, P, r, \gamma\}$, with finite state and action spaces.

The advantage function is defined by:

$$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$$

The intuition for this function is that $A^\pi(s,a)$ can be considered the improvement over $V^\pi(s)$ when taking action a. d

4.1 Q and V function relationship [5 points]

Consider a stochastic policy $\pi(a|s)$. Write down an equality that expresses $V^\pi(s)$ only in terms of $Q^\pi(s,a)$ and $\pi(a|s)$.

4.2 The Expected Advantage [10 points]

Compute the value of $E_{a \sim \pi(\cdot|s)}[A^\pi(s,a)]$. Your answer should hold for all states s and stochastic policies π.

4.3 Policy Iterations in terms of the Advantages [10 points]

Write out the policy iteration algorithm in terms of only $A^{\pi}(s,a)$ rather than $Q^{\pi}(s,a)$. Your algorithm should produce the same sequence of intermediate policies as the standard PI algorithm, as specified in terms of $Q^{\pi}(s,a)$. Justify your answer.

4.4 Characterizing $\pi^\star$ in terms of the Advantages [10 points]

Suppose π is a deterministic function. Write down an inequality stated only in terms of $A^\pi(s, a)$ such that if this inequality holds, then it must be the case that $\pi = \pi^\star$. Your inequality may be stated in terms of $A^\pi(s, a)$ and $\pi(s)$, where s and a are free variables. Justify your answer using the Bellman equations for V^π.

5 The Cubic Quadratic Regulator [30 points]

Consider the following control problem with 1-dimensional state x and control u, horizon H, cost function $c(x, u) = x^2 + u^2$, and nonlinear (deterministic) dynamics

$$f(x, u) = x^3 + u^3.$$

Recall the local linearization approach covered in class that converts a nonlinear control problem with dynamics f and cost c into a linear control problem with dynamics $\tilde{f}$ and cost $\tilde{c}$, where $\tilde{f}$ and $\tilde{c}$ are the linearization and quadratization of f and c respectively about the target point $(0, 0)$.

5.1 Local Linearization Failure [15 points]

What goes wrong if we apply local linearization to the above nonlinear control problem? Assume that we have access to exact derivatives and do not need to use the finite differencing approximation covered in class.

5.2 Local Linearization Success [15 points]

1. Propose an alternative local-linearization-based method that converts f, c, and H into an LQR problem (with stationary dynamics and cost function) and outputs the optimal controller for that problem.
2. Explain intuitively why your algorithm returns a better solution to the specific control problem above. (You do not need to provide a rigorous proof.)

6 Variations of Thompson Sampling [35 points]

Consider a K-armed bandit. Recall Thompson Sampling, at each time in a bandit, takes a draw $\boldsymbol{\mu} := (\mu_1, \ldots, \mu_K)$ from the posterior of the arm means and takes the optimal action with respect to that draw, i.e., $\arg\max_k \mu_k$. Equivalently, it takes a draw $k^\star$ from the posterior of the index of the optimal arm, and then simply takes that draw as its action, i.e., it takes action $k^\star$.

6.1 Thompson sampling with the posterior mean [5 points]

Consider a version of Thompson Sampling that instead takes the optimal action with respect to the posterior mean, i.e., $\arg\max_k \bar{\mu}_k$, where $\bar{\mu}_k$ denotes the current time point's posterior mean of arm k.

Is this algorithm more or less greedy than standard Thompson Sampling? Will this algorithm achieve sublinear expected regret? Explain your answer.

6.2 Thompson sampling with two $\boldsymbol{\mu}$ samples [10 points]

Consider a version of Thompson Sampling that, at each time, takes two draws $\boldsymbol{\mu}_a$ and $\boldsymbol{\mu}_b$ from the posterior, computes their mean, $\tilde{\boldsymbol{\mu}} = \frac{\boldsymbol{\mu}_a + \boldsymbol{\mu}_b}{2}$, and then takes the action that is optimal with respect to $\tilde{\boldsymbol{\mu}}$.

Is this algorithm more or less greedy than standard Thompson Sampling? Will this algorithm achieve sublinear expected regret? Explain your answer.

6.3 Thompson Sampling with two $k^\star$ samples [10 points]

Consider a version of Thompson Sampling that, at each time, instead of taking the action which is a draw from the posterior of $k^\star$ (the optimal arm), it takes two independent samples from this posterior, k_a and k_b, and takes the action corresponding to the sample with higher posterior mean, i.e., $\arg\max_{k \in \{k_a, k_b\}} \bar{\mu}_k$.

Is this algorithm more or less greedy than standard Thompson Sampling? Will this algorithm achieve sublinear expected regret? Explain your answer.

6.4 Thompson Sampling with two $k^\star$ samples, version 2 [10 points]

Continuing with the setup of the previous question (with two draws k_a and k_b), consider the algorithm that takes the action corresponding to the sample with lower posterior mean, i.e., $\arg\min_{k\in\{k_a,k_b\}} \bar{\mu}_k$.

Is this algorithm more or less greedy than standard Thompson Sampling? Will this algorithm achieve sublinear expected regret? Explain your answer.